

SOLWEIG 2022a — Execution Flow & Formula Reference

Scope: Full technical trace from QGIS plugin entry point through every computational stage to the final Tmrt raster.

Version tag: `Solweig_2022a_calc` (active as of commit `e0c3d5a`).

Key files: `SOLWEIG/solweig.py`, `SOLWEIG/solweigworker.py`, `SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py` and helper modules.

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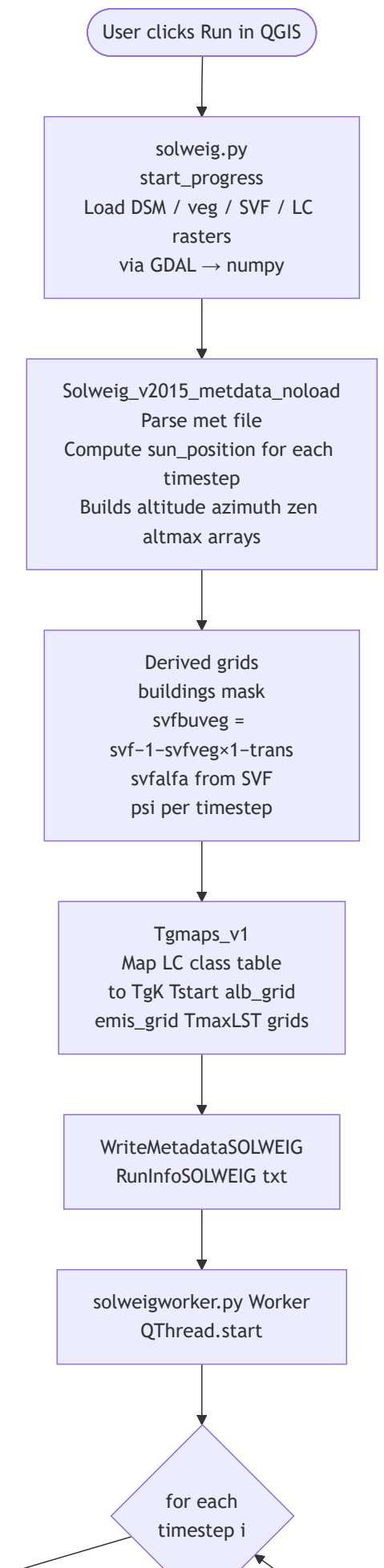
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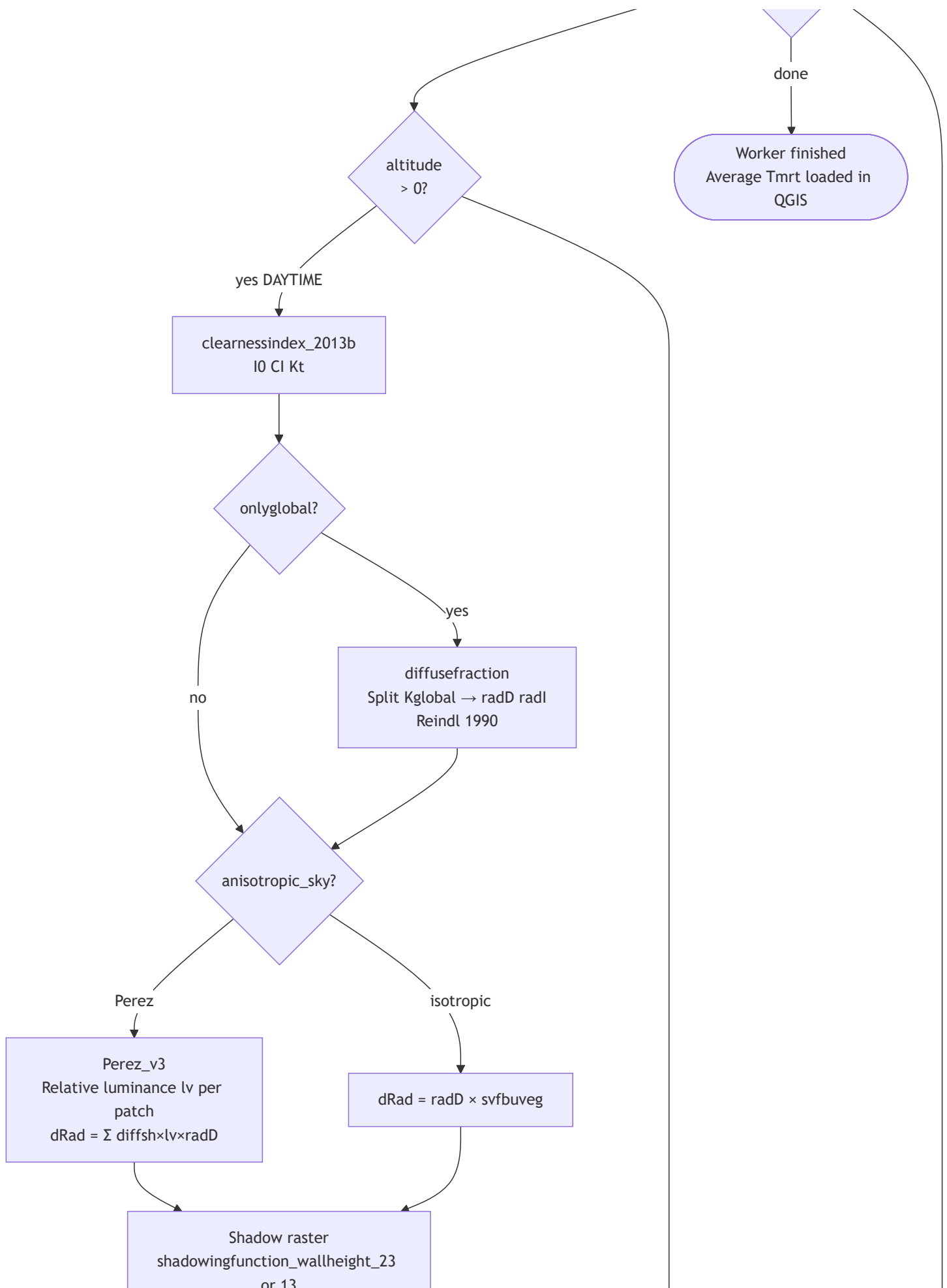
1. Execution Flow — Ordered Steps

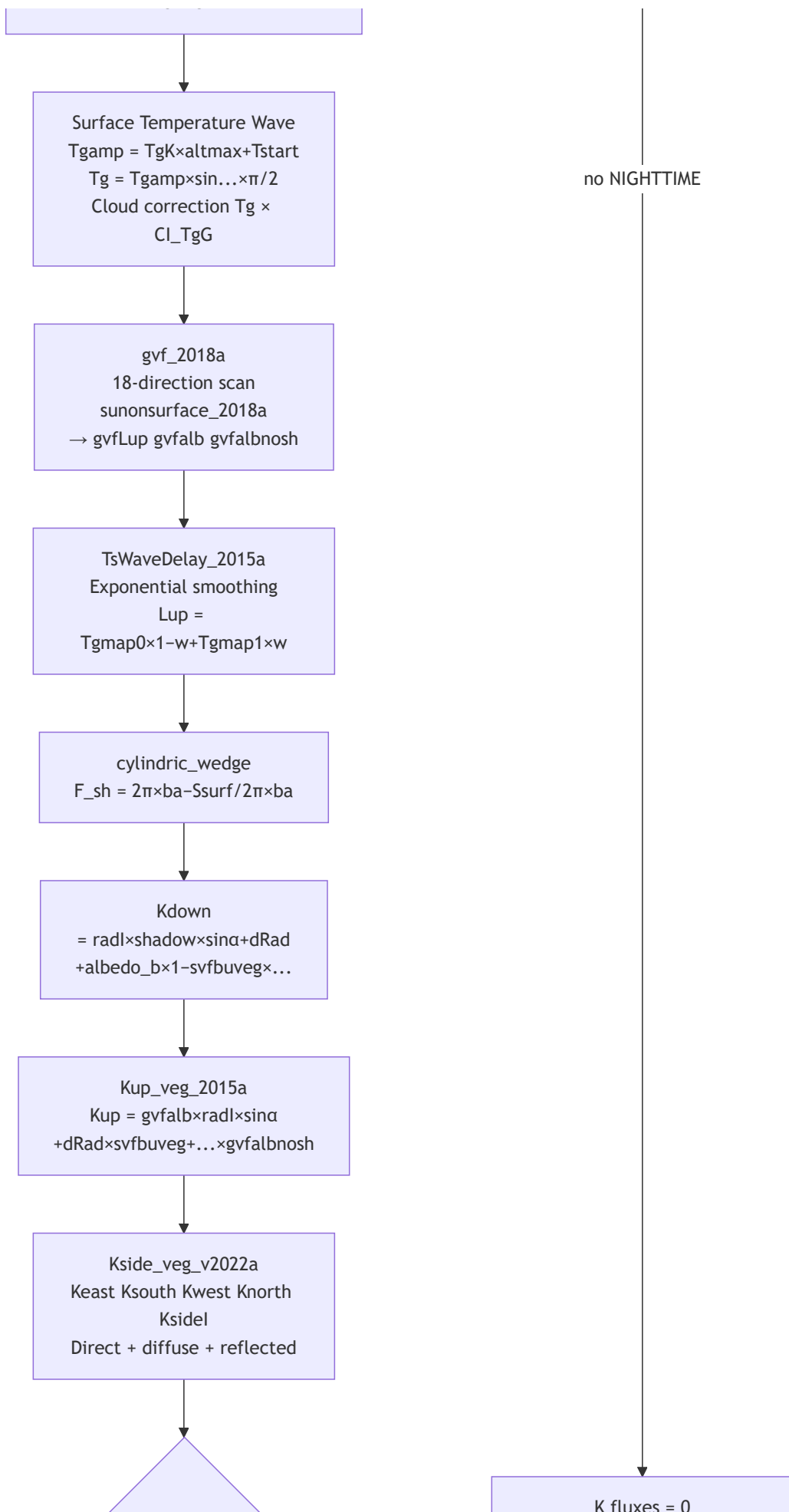
#	Module / Function	Purpose	Key Inputs	Key Outputs
1	<code>solweig.py</code> → <code>SOLWEIG.__init__</code>	Plugin is registered with QGIS. UI is wired.	QGIS plugin API	Plugin object, menu entry
2	<code>solweig.py</code> → <code>SOLWEIG.run</code>	Dialog opens. Combo-boxes for rasters are populated from QGIS layer registry.	QGIS layers, previously saved settings	Populated <code>SOLWEIGDialog</code>
3	<code>solweig.py</code> → <code>SOLWEIG.start_progress</code>	User clicks Run. All inputs are validated and spatial rasters are loaded via GDAL into numpy arrays.	DSM, DEM, vegDSM×2, wall height, wall aspect, SVF zip, LC raster, met file path, POI vector layer	<code>dsm</code> , <code>dem</code> , <code>vegdsm</code> , <code>svf*</code> (19 arrays), <code>lcgrid</code> , <code>walls</code> , <code>dirwalls</code> , <code>metdata</code> (n×24 array)
4	<code>Solweig_v2015_metdata_noload.py</code> → <code>Solweig_2015a_metdata_noload</code>	Parses the met file; calls <code>sun_position</code> for every timestep (+ daily 15-min scan for <code>altmax</code>).	<code>metdata</code> array, location dict, UTC offset	<code>altitude[1,n]</code> , <code>azimuth[1,n]</code> , <code>zen[1,n]</code> (rad), <code>jday[1,n]</code> , <code>dectime[n]</code> , <code>altmax[1,n]</code> , <code>leafon[1,n]</code>
5	<code>solweig.py</code> → derived grids	Builds <code>buildings</code> mask (DSM-DEM or LC), <code>svfbuveg</code> (effective sky fraction with vegetation transmissivity), <code>svfalfa</code> (SVF-weighted building-height angle), <code>psi</code> (vegetation transmissivity vector per timestep).	<code>svf</code> , <code>svfveg</code> , <code>leafon</code> , UI transmissivity controls	<code>buildings</code> , <code>svfbuveg</code> , <code>svfalfa</code> , <code>psi[1,n]</code>
6	<code>Tgmaps_v1.py</code> → <code>Tgmaps_v1</code>	Maps LC class table columns to per-pixel grids.	<code>lc_grid</code> , <code>lc_class</code> (from <code>landcoverclasses_2016a.txt</code>)	<code>TgK</code> , <code>Tstart</code> , <code>alb_grid</code> , <code>emis_grid</code> , <code>TmaxLST</code> , <code>TgK_wall</code> , <code>Tstart_wall</code> , <code>TmaxLST_wall</code>
7	<code>WriteMetadataSOLWEIG.py</code> → <code>writeRunInfo</code>	Writes <code>RunInfoSOLWEIG*.txt</code> with all run parameters.	Run configuration	Metadata text file
8	<code>solweigworker.py</code> → <code>Worker.__init__</code>	Worker object is constructed with ~80 parameters;	All of the above	Worker object

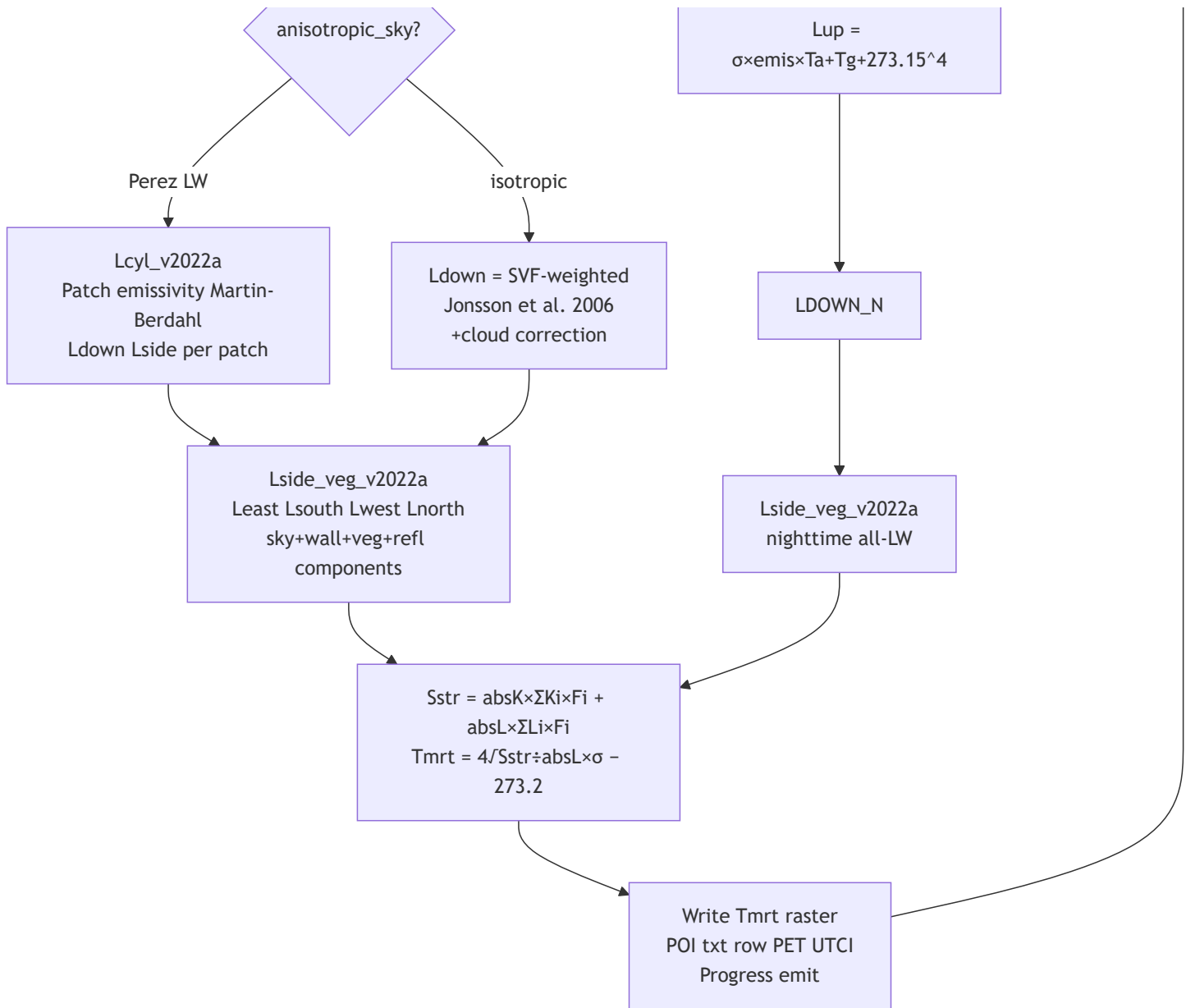
#	Module / Function	Purpose	Key Inputs	Key Outputs
		QThread.start() is called.		
9	solweigworker.py → Worker.run — time loop	Iterates $i = 0 \dots N-1$ over all meteorological timesteps. At midnight (dectime mod 1 == 0) refreshes <code>Twater</code> and looks ahead for the first sunlit CI value.	T_a , RH , $radG$, $radD$, $radI$, P , W_s time series	Per-timestep output arrays passed to step 10
10	Solweig_2022a_calc.py → Solweig_2022a_calc (daytime branch)	Full per-timestep radiation physics (see §3 for formulas).	One row of met data + all spatial grids	T_{mrt} , K_{down} , K_{up} , L_{down} , L_{up} , $K_{east/S/W/N}$, $L_{east/S/W/N}$, $shadow$, T_g , $esky$, CI , ...
10N	Solweig_2022a_calc.py → Solweig_2022a_calc (nighttime branch)	All shortwave fluxes = 0; longwave uses nocturnal Stefan–Boltzmann with T_a+T_g .	Same as 10 but $altitude \leq 0$	Same return tuple with K-fluxes = 0
11	solweigworker.py → POI writing	Samples grid arrays at POI pixel locations; computes PET and UTCI; appends 35-column row to <code>POI_<name>.txt</code> .	Grid arrays, $poisxy$, W_s , person parameters	<code>POI_*.txt</code> rows (35 cols)
12	solweigworker.py → raster writing	Writes selected GeoTIFFs (T_{mrt} , $Shadow$, K_{down} , K_{up} , L_{down} , L_{up} , etc.) for this timestep.	Grid arrays, <code>gdal_dsm</code> georef	<code>Tmrt_YYYY_DOY_HHMMw.tif</code> , etc.
13	solweig.py → finishedWorker	Receives average <code>tmrtpplot</code> from worker; loads it as a temporary QGIS raster layer.	<code>tmrtpplot/N</code>	QGIS canvas layer

2. Full Pipeline Flowchart (Mermaid)









3. Formulas Along the Execution Path

Notation used throughout:

- $\sigma = 5.67051 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ (Stefan–Boltzmann constant)
- T_a = air temperature (°C); converted to K as $T_a + 273.15$
- Angles in degrees unless explicitly noted as radians

F-1 Solar geometry and declination

File: SOLWEIG/SOLWEIGpython/daylen.py lines 12–19

Also used in: Utilities/SEBESOLWEIGCommonFiles/Solweig_v2015_metdata_noload.py

Solar declination

$$\delta = -23.45 \cos\left(\frac{2\pi (DOY + 10)}{365}\right)$$

Symbol	Meaning	Unit
δ	Solar declination	°
DOY	Day of year (1–365/366)	—

Sunrise time

$$SOC = \tan(\delta \cdot RAD) \cdot \tan(\phi \cdot RAD)$$

$$DAYL = 12 + \frac{24}{\pi} \arcsin(SOC), \quad SNUP = 12 - \frac{DAYL}{2}$$

Symbol	Meaning	Unit
SOC	Sine of critical angle (clamped ±1)	—
ϕ	Geographic latitude	°
$DAYL$	Day length	h
$SNUP$	Decimal hour of sunrise	h

Constant: $RAD = \pi/180$

Sun altitude α , azimuth, and zenith θ_z at each timestep are computed by `sun_position` (separate module, not reproduced here). `altmax` is found by 15-min stepping through the day to locate the maximum elevation angle.

F-2 Vapor pressure

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py line 95

$$e_a = 6.107 \times 10^{\frac{7.5 T_a}{237.3 + T_a}} \times \frac{RH}{100}$$

Symbol	Meaning	Unit
e_a	Actual vapor pressure	hPa
T_a	Air temperature	°C
RH	Relative humidity	%

F-3 Atmospheric (sky) emissivity — Prata 1996

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py lines 98–99

$$\xi = 46.5 \cdot \frac{e_a}{T_a + 273.15}$$

$$\varepsilon_{sky} = 1 - (1 + \xi) \exp\left(-\sqrt{1.2 + 3\xi}\right) + \Delta$$

Symbol	Meaning	Unit
ξ	msteg — reduced precipitable water depth	—
e_a	Vapor pressure (F-2)	hPa
ε_{sky}	Clear-sky atmospheric emissivity	—
Δ	elvis correction flag (0 or 1, UI controlled)	—

Note: The original Jonsson et al. (2006) implementation had an erroneous −0.04 offset; removed in 2022a.

F-4 Clear-sky direct normal irradiance and clearness index

File: Utilities/SEBESOLWEIGCommonFiles/clearnessindex_2013b.py lines 29–83

Caller: Solweig_2022a_calc.py line 104

Optical air mass (Kasten & Young)

$$m = \frac{35 \cos \theta_z}{(1224 \cos^2 \theta_z + 1)^{1/2}}$$

Transmission coefficients

$$T_{rpg} = 1.021 - 0.084 \left[m \left(0.000949 p + 0.051 \right) \right]^{0.5}$$

$$T_w = 1 - 0.077 \left(u m \right)^{0.3}$$

$$T_{ar} = 0.935^m$$

where $u = \exp(0.1133 - \ln(G + 1) + 0.0393 T_d)$ is the precipitable water (cm), T_d the dewpoint in °F, and G a latitude- and season-dependent empirical constant (tabulated in code, lines 33–59).

Clear-sky horizontal irradiance

$$I_0 = I_{TOA} \cdot \cos \theta_z \cdot T_{rpg} \cdot T_w \cdot D \cdot T_{ar}$$

Symbol	Meaning	Unit
I_0	Clear-sky global on horizontal	W m ^{−2}
I_{TOA}	Solar constant = 1370	W m ^{−2}
θ_z	Solar zenith angle	rad
D	Sun–Earth distance correction factor	—
p	Station pressure	mbar

Low-sun-elevation correction (Lindberg et al. 2008)

$$corr = 0.1473 \ln(90 - \theta_{z,deg}) + 0.3454$$

Clearness Index

$$CI = \frac{K_{global}}{I_0} + (1 - corr), \quad CI \leq 1$$

Symbol	Meaning	Unit
CI	Clearness index (cloud cover proxy)	0–1
K_{global}	Measured global shortwave	W m ⁻²

F-5 Diffuse / direct shortwave splitting — Reindl et al. 1990

File: Utilities/SEBESOLWEIGCommonFiles/diffusefraction.py lines 28–35

Caller: Solweig_2022a_calc.py line 114 (when onlyglobal == 1)

$$K_t = \frac{K_{global}}{I_{0,et}}, \quad I_{0,et} = I_{TOA} \cos \theta_z \cdot D$$

Diffuse fraction (three piecewise regimes):

$$\frac{K_{diff}}{K_{global}} = \begin{cases} 1 - 0.232K_t + 0.0239 \sin \alpha - 0.000682T_a + 0.0195 \frac{RH}{100} & K_t \leq 0.3 \\ 1.329 - 1.716K_t + 0.267 \sin \alpha - 0.00357T_a + 0.106 \frac{RH}{100} & 0.3 < K_t < 0.78 \\ 0.426K_t - 0.256 \sin \alpha + 0.00349T_a + 0.0734 \frac{RH}{100} & K_t \geq 0.78 \end{cases}$$

Direct beam normal irradiance:

$$K_{dir} = \frac{K_{global} - K_{diff}}{\sin \alpha}$$

Symbol	Meaning	Unit
α	Solar altitude angle	°
K_t	Clearness ratio (vs. extraterrestrial)	—
K_{diff}	Diffuse horizontal irradiance	W m ⁻²
K_{dir}	Direct normal irradiance	W m ⁻²

F-6 Surface temperature diurnal wave

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py lines 149–153

Fixed in 2021a (old formula had double-subtraction of Tstart)

Temperature wave amplitude

$$T_{g,amp} = T_{gK} \cdot \alpha_{max} + T_{start}$$

$$T_{g,amp,wall} = T_{gK,wall} \cdot \alpha_{max} + T_{start,wall}$$

Diurnal wave (sinusoidal)

$$T_g = T_{g,amp} \cdot \sin\left(\frac{(t_{dec} - SNUP/24)}{(T_{maxLST}/24 - SNUP/24)} \cdot \frac{\pi}{2}\right)$$

where $t_{dec} - \lfloor t_{dec} \rfloor$ is the fractional day (i.e., time-of-day in decimal days).

Symbol	Meaning	Unit
T_g	Ground surface temperature excess above T_a	°C
T_{gK}	Amplitude coefficient (from LC class table, col 3)	°C per degree of solar altitude
α_{max}	Maximum solar altitude on that day	°
T_{start}	Minimum surface temperature offset (col 4)	°C
t_{dec}	Decimal time (DOY + fractional hour)	days
T_{maxLST}	Hour of maximum surface temperature (col 5)	h
$SNUP$	Sunrise hour (from F-1)	h

Land cover defaults (from landcoverclasses_2016a.txt):

Class	T_{gK}	T_{start}	T_{maxLST}	albedo	emissivity
1 Dark asphalt	0.37	3.0	15.0	0.18	0.95
2 Roofs	0.37	3.0	15.0	0.18	0.95
5 Grass	0.25	1.5	15.0	0.16	0.94
6 Bare soil	0.30	2.0	15.0	0.25	0.94
7 Water	0.05	0.5	15.0	0.05	0.98
99 Walls	0.37	3.0	15.0	0.20	0.90

F-7 Cloud correction for surface temperature

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py lines 161–175

$$radI_0, radD_{,I_0} = diffusefraction(I_0, \alpha, 1.0, T_a, RH)$$
$$radG_0 = radI_0 \cdot \sin \alpha + radD_{,I_0}$$
$$CI_{T_gG} = \frac{K_{global}}{radG_0} + (1 - corr), \quad CI_{T_gG} \leq 1$$
$$T_g \leftarrow T_g \cdot CI_{T_gG}$$

Symbol	Meaning
$radG_0$	Clear-sky global radiation (derived from I_0)
CI_{T_gG}	Cloudiness reduction factor for surface temperature

Derived from code — not referenced to a single external publication at this line.

F-8 Vegetation shadow attenuation

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py lines 137–144

$$shadow = sh - (1 - vegsh)(1 - \psi)$$

Symbol	Meaning	Unit
sh	Binary shadow raster from building-only ray-casting	0 (shaded) / 1 (sunlit)
$vegsh$	Vegetation shadow fraction raster	0–1
ψ	Shortwave vegetation transmissivity (<code>1 - transmissivity</code> used as opacity)	0–1
$shadow$	Effective shadow accounting for vegetation transmissivity	0–1

Note: `psi` in code comments is defined as `1 - transmissivity`. The shadow formula therefore partially restores radiation where vegetation is present but semi-transparent.

F-9 Anisotropic diffuse irradiance — Perez et al. 1993

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py lines 118–128

Called helper: Utilities/SEBESOLWEIGCommonFiles/Perez_v3.py

$$dRad = \left(\sum_{p=1}^{N_{patch}} diffsh_{:,:,p} \cdot lv_{p,2} \right) \cdot K_{diff}$$

Symbol	Meaning
$diffsh_{:,:,p}$	Pre-computed fraction of patch p visible from each cell (from <code>shadowmats.npz</code>)
$lv_{p,2}$	Relative luminance of patch p from Perez model (normalized to integrate to 1)
K_{diff}	Diffuse horizontal irradiance (W m ⁻²)
$dRad$	Spatially distributed diffuse shortwave (W m ⁻²)

Isotropic fallback (when `anisotropic_sky == 0`):

$$dRad = K_{diff} \cdot svfbuveg$$

where $svfbuveg = svf - (1 - svfveg)(1 - trans)$ is the effective sky fraction accounting for vegetation.

F-10 Cylindric wedge — F_sh (wall shadow fraction)

File: SOLWEIG/SOLWEIGpython/cylindric_wedge.py lines 17–44

This function computes the fraction of the vertical surface of the cylinder (standing person model) that is shaded by surrounding buildings, as a function of the sun zenith angle and SVF-derived building-height angle.

Let $\beta = \theta_z$ (zenith, rad), $\alpha = svfalfa$ (SVF building-height angle, rad):

$$x_a = 1 - \frac{2}{\tan \alpha \cdot \tan \beta}, \quad h_a = \frac{2}{\tan \alpha \cdot \tan \beta}, \quad b_a = \frac{1}{\tan \alpha}$$

When $x_a < 0$ (sun low enough to be partially blocked):

$$q_a = \frac{\tan \beta}{2}, \quad Z_a = \sqrt{b_a^2 - q_a^2/4}$$

$$\phi = \arctan(Z_a/q_a), \quad A = \frac{\sin \phi - \phi \cos \phi}{1 - \cos \phi}$$

$$S_{surf} = 2b_a h_a + 2b_a x_a A$$

$$F_{sh} = \frac{2\pi b_a - S_{surf}}{2\pi b_a}$$

When $x_a \geq 0$: $F_{sh} = 0$ (fully sunlit cylinder wall).

Symbol	Meaning
F_{sh}	Fraction of cylinder wall in shadow (0 = fully lit, 1 = fully shaded)
$svfalfa$	SVF-derived effective building-height angle (rad)

F-11 Ground View Factor upwelling longwave (Lup component per direction)

File: SOLWEIG/SOLWEIGpython/sunonsurface_2018a.py lines 22–26

Caller: gv_f_2018a.py via 18-direction azimuth scan

Per-pixel upwelling longwave excess above the air term:

$$L_{up,excess} = \sigma \varepsilon_{surf} [(T_g \cdot shadow + T_a + 273.15)^4 - (T_a + 273.15)^4]$$

Sunlit wall excess:

$$L_{wall} = \sigma \varepsilon_{wall} [(T_{g,wall} + T_a + 273.15)^4 - (T_a + 273.15)^4]$$

Symbol	Meaning	Unit
ε_{surf}	emis_grid — surface emissivity per pixel	—
ε_{wall}	ewall — wall emissivity (default 0.9)	—
$T_g \cdot shadow$	Ground temperature excess (zero in shade)	°C

GVF weighting (sunonsurface_2018a.py line 194): Two-distance weights (near = 0– first , far = 0– second) combined as:

$$GVF = \frac{0.5 \cdot GV F_1 + 0.4 \cdot GV F_2}{0.9}$$

Derived from code — weighting constants 0.5 / 0.4 are empirical, not published.

F-12 GVF directional averaging and background air term

File: SOLWEIG/SOLWEIGpython/gvf_2018a.py lines 64–83

18 azimuth directions $A = \{5^\circ, 25^\circ, 45^\circ, \dots, 355^\circ\}$ (step 20°):

$$gvfLup = \frac{1}{18} \sum_{j=1}^{18} gvfLup_j + \sigma \varepsilon_{surf} (T_a + 273.15)^4$$

Cardinal half-hemisphere averages (9 directions each):

$$gvfLup_E = \frac{1}{9} \sum_{j:A_j \in [0^\circ, 180^\circ)} gvfLup_j + \sigma \varepsilon_{surf} (T_a + 273.15)^4$$

(same for S, W, N with corresponding azimuth windows)

F-13 Surface temperature wave-delay smoothing

File: SOLWEIG/SOLWEIGpython/TsWaveDelay_2015a.py lines 11–21

$$w_1 = \exp(-33.27 \cdot t_{add})$$

$$L_{up,smooth} = L_{up,current} \cdot (1 - w_1) + L_{up,prev} \cdot w_1$$

Symbol	Meaning	Unit
w_1	Exponential decay weight	—
t_{add}	Accumulated time since last update	decimal days
$L_{up,current}$	gvfLup from current timestep	W m ⁻²
$L_{up,prev}$	Tgmap1 — stored previous smoothed value	W m ⁻²

Constants: decay coefficient = 33.27 days⁻¹; update threshold ≥ 59 min.

Coefficient is empirical — source publication not cited in code comments.

F-14 Downward shortwave on horizontal plane (Kdown)

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py lines 202–203

$$K_{down} = K_{dir} \cdot shadow \cdot \sin \alpha + dRad + \alpha_b \cdot (1 - svfbuveg) \cdot [K_{global}(1 - F_{sh}) + K_{diff} F_{sh}]$$

Symbol	Meaning	Unit
K_{dir}	Direct normal irradiance	W m ⁻²
$shadow$	Effective shadow fraction (F-8)	0–1
α	Solar altitude	°
$dRad$	Diffuse sky radiation reaching cell (F-9)	W m ⁻²
α_b	albedo_b — wall/building albedo	—
$svfbuveg$	Effective SVF with vegetation	0–1
F_{sh}	Cylindric wedge shadow fraction (F-10)	0–1

The third term accounts for shortwave reflected off the surrounding building walls and vegetation toward the point of interest.

F-15 Upward shortwave (Kup) from ground and walls

File: SOLWEIG/SOLWEIGpython/Kup_veg_2015a.py lines 5–13

$$K_{up} = gvfabl \cdot K_{dir} \cdot \sin \alpha + [K_{diff} svfbuveg + \alpha_b(1 - svfbuveg)(K_{global}(1 - F_{sh}) + K_{diff}F_{sh})] \cdot gvfablnosh$$

Symbol	Meaning
$gvfabl$	GVF-weighted albedo of sunlit surfaces seen from the point
$gvfablnosh$	GVF-weighted albedo of all surfaces regardless of shadow state

Cardinal directions $K_{up,E/S/W/N}$ use the same formula with directional $gvfabl^*$ and $gvfablnosh^*$ arrays from `gvf_2018a`.

F-16 Shortwave on vertical cylinder sides (KsideI / Kside)

File: SOLWEIG/SOLWEIGpython/Kside_veg_v2022a.py line 46

Direct beam on cylinder (integrated over azimuth, perpendicular projection):

$$K_{side,I} = shadow \cdot K_{dir} \cdot \cos \alpha$$

Anisotropic diffuse on cylinder (loop over Perez patches):

$$K_{side,D} = \sum_p diffsh_{:,p} \cdot \frac{K_{diff} lv_{p,2}}{radTot} \cdot \cos \alpha_p \cdot \Omega_p$$

where α_p is the patch altitude, Ω_p is the steradian of patch p .

Reflected shortwave contributions (anisotropic, cylinder):

$$K_{ref,veg} = \sum_p \frac{\alpha_b \cdot K_{diff} \cdot 0.5}{\pi} \cdot mask_{veg,p} \cdot \Omega_p \cdot \cos \alpha_p$$

$$K_{ref,sun/sh} = \sum_p \frac{\alpha_b \cdot (K_{dir} \cos \alpha + K_{diff} \cdot 0.5) \text{ or } K_{diff} \cdot 0.5}{\pi} \cdot mask_{sun/sh,p} \cdot \Omega_p \cdot \cos \alpha_p$$

Total:

$$K_{side} = K_{side,I} + K_{side,D} + K_{ref,sun} + K_{ref,sh} + K_{ref,veg}$$

F-17 Shortwave on box cardinal faces (Keast/Ksouth/Kwest/Knorth)

File: SOLWEIG/SOLWEIGpython/Kside_veg_v2022a.py lines 49–63 (isotropic), lines 155–232 (anisotropic)

Direct on east face (box model, isotropic sky):

$$K_{east,I} = K_{dir} \cdot shadow \cdot \cos \alpha \cdot \sin(azimuth + t), \quad \text{if } azimuth \in (360^\circ - t, 180^\circ - t]$$

(south: $\sin(azimuth - 90^\circ + t)$; west: $\sin(azimuth - 180^\circ + t)$; north: $\sin(azimuth - 270^\circ + t)$)

Diffuse + reflected on east face (isotropic):

$$K_{east,DG} = \frac{1}{2} \left[K_{diff}(1 - svfvikt_E) + \alpha_b \cdot svfvikt_E \cdot (K_{global}(1 - F_{sh}) + K_{diff}F_{sh}) + K_{up,E} \right]$$

where the SVF weight for sidewall vegetation is:

$$svfvikt_E = \frac{vikttot - P_6(svf_{veg+bu,E})}{vikttot}$$

with P_6 a 6th-degree polynomial of the combined vegetation+building SVF (from kvikt_veg.py lines 5–8):

$$P_6(x) = 63.227x^6 - 161.51x^5 + 156.91x^4 - 70.424x^3 + 16.773x^2 - 0.4863x$$

Symbol	Meaning
$vikttot = 4.4897$	Total weight of the SVF polynomial (empirical constant)
$svfvikt_E$	Fraction of radiation blocked by walls + vegetation towards east

F-18 SVF-weighted wall-height angle

File: SOLWEIG/SOLWEIGpython/Lside_veg_v2022a.py lines 10–13

$$svfalfa_{dir} = \arcsin \left(\exp \left(\frac{\ln(1 - svf_{dir})}{2} \right) \right)$$

This inverts the SVF formula $svf = 1 - \sin^2(\alpha_{wall})$ to recover a representative wall-height angle for each cardinal direction.

F-19 Isotropic downward longwave (Ldown) — Jonsson et al. 2006

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py lines 293–295

$$\begin{aligned} L_{down} = & (svf + svfveg - 1) \varepsilon_{sky} \sigma (T_a + 273.15)^4 \\ & + (2 - svfveg - svfaveg) \varepsilon_{wall} \sigma (T_a + 273.15)^4 \\ & + (svfaveg - svf) \varepsilon_{wall} \sigma (T_a + 273.15 + T_{g,wall})^4 \\ & + (2 - svf - svfveg) (1 - \varepsilon_{wall}) \varepsilon_{sky} \sigma (T_a + 273.15)^4 \end{aligned}$$

Term	Physical meaning
Term 1	Sky longwave visible through combined building+veg sky fraction
Term 2	Longwave from wall pixels at air temperature (shaded walls or walls not in sun angle)
Term 3	Longwave from sunlit walls elevated by $T_{g,wall}$
Term 4	Specular reflection of sky radiation off walls

Symbol	Meaning	Unit
svf	Sky view factor (buildings only)	0–1
$svfveg$	Sky view factor (vegetation blocking sky)	0–1
$svfaveg$	Vegetation SVF blocking buildings	0–1
ϵ_{wall}	ewall wall emissivity	—
$T_{g,wall}$	Wall temperature excess (F-6)	°C

F-20 Cloudy-sky correction for isotropic Ldown

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py lines 301–305 (isotropic) and lines 284–286 (anisotropic)

Anisotropic path ($CI < 0.95$):

$$\epsilon_{sky,c} = CI \cdot \epsilon_{sky} + (1 - CI) \cdot 1$$

i.e., blend between clear-sky and overcast-sky ($\epsilon = 1$) emissivity.

Isotropic path ($c = 1 - CI$):

$$L_{down,cloudy} = L_{down}(1 - c) + c \cdot L_{down,\epsilon_{sky}=1}$$

where $L_{down,\epsilon_{sky}=1}$ uses $\epsilon_{sky} = 1$ (blackbody cloud cover).

F-21 Anisotropic sky emissivity per patch — Martin & Berdahl 1984

File: SOLWEIG/SOLWEIGpython/emissivity_models.py lines 76

Caller: Lcyl_v2022a.py line 37 (hardcoded emis_m = 2)

$$\epsilon_{sky}(\theta_z) = 1 - (1 - \epsilon_{sky}) \cdot \exp\left[b_c \left(1.7 - \frac{1}{\cos \theta_z}\right)\right]$$

Symbol	Meaning	Unit
ϵ_{sky}	Hemispherically averaged sky emissivity (F-3)	—
$\epsilon_{sky}(\theta_z)$	esky_band — emissivity at zenith angle θ_z of each sky patch	—
b_c	Empirical constant = 0.308 (Ångström 1915, via Nahon et al. 2019)	—

F-22 Anisotropic Ldown and Lside on cylinder (Lcyl_v2022a)

File: SOLWEIG/SOLWEIGpython/Lcyl_v2022a.py lines 70–72

Longwave flux from each sky patch p at altitude α_p , steradian Ω_p :

On horizontal surface:

$$L_{down,p} = \frac{\epsilon_{sky,p} \sigma (T_a + 273.15)^4}{\pi} \cdot \Omega_p \cdot \sin \alpha_p$$

On vertical surface (side of cylinder):

$$L_{side,p} = \frac{\varepsilon_{sky,p} \sigma (T_a + 273.15)^4}{\pi} \cdot \Omega_p \cdot \cos \alpha_p$$

Perpendicular (normal to patch):

$$L_{normal,p} = \frac{\varepsilon_{sky,p} \sigma (T_a + 273.15)^4}{\pi} \cdot \Omega_p$$

Then `patch_characteristics.define_patch_characteristics` weights each patch by its shadow/vegetation/building type (sky, sunlit wall, shaded wall, ground) to produce total L_{down} and L_{side} grids.

F-23 Longwave from cardinal-direction walls (Lside_veg_v2022a)

File: SOLWEIG/SOLWEIGpython/Lside_veg_v2022a.py

All-sky correction:

$$L_{sky,allsky} = \varepsilon_{sky} \sigma (T_a + 273.15)^4 \cdot (1 - c) + c \sigma (T_a + 273.15)^4, \quad c = 1 - CI$$

Isotropic sky contribution to east face (line 50):

$$L_{sky,E} = \frac{(svf_E + svf_{veg_E} - 1) \cdot L_{sky,allsky} \cdot vikt_{s_E}}{2}$$

Sunlit wall contribution (lines 35–36, example east, daytime):

$$L_{wall,sun,E} = \sigma \varepsilon_{wall} (T_a + 273.15 + T_w \sin(azi_E))^4 \cdot vikt_{wall_E} \cdot (1 - F_{sh}) \cdot \cos \beta_{sun} \cdot 0.5$$

Shaded wall:

$$L_{wall,sh,E} = \sigma \varepsilon_{wall} (T_a + 273.15)^4 \cdot vikt_{wall_E} \cdot F_{sh} \cdot 0.5$$

Vegetation:

$$L_{veg,E} = \sigma \varepsilon_{wall} (T_a + 273.15)^4 \cdot vikt_{veg_E} \cdot 0.5$$

Ground (upward longwave from that quadrant):

$$L_{ground,E} = L_{up,E} \cdot 0.5$$

Reflected:

$$L_{refl,E} = (L_{down} + L_{up,E}) \cdot vikt_{refl_E} \cdot (1 - \varepsilon_{wall}) \cdot 0.5$$

Total isotropic east:

$$L_{east} = L_{sky,E} + L_{wall,sun,E} + L_{wall,sh,E} + L_{veg,E} + L_{ground,E} + L_{refl,E}$$

When anisotropic longwave is active, only $L_{ground,E} = L_{up,E} \cdot 0.5$ is returned; sky/wall contributions come from `Lcy1_v2022a`.

Symbol	Meaning
T_w	$T_{g,wall}$ — wall surface temperature excess (°C)
β_{sun}	Effective sun angle on wall derived from SVF building-height angle and F_{sh}

Symbol	Meaning
$vikts_E, viktwall_E, viktveg_E, viktrefl_E$	SVF-polynomial weights for sky / walls / vegetation / reflected longwave towards east

F-24 Nighttime upwelling longwave

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py line 247

$$L_{up} = \sigma \varepsilon_{surf} (T_a + T_g + 273.15)^4$$

During nighttime $T_g = 0$ (all K-fluxes zero), so this reduces to:

$$L_{up,night} = \sigma \varepsilon_{surf} (T_a + 273.15)^4$$

Water bodies (LC class 3, line 249): $L_{up} = \sigma \cdot 0.98 \cdot (T_{water} + 273.15)^4$, where T_{water} = daily mean air temperature.

F-25 Mean Radiant Temperature (Tmrt)

File: SOLWEIG/SOLWEIGpython/Solweig_2022a_calc.py lines 321–334

Angular factors set in: SOLWEIG/solweig.py lines 718–727

Absorbed radiant flux density

Three model variants depending on posture (cyl) and sky model (anisotropic_sky):

Case A — Cylinder, isotropic sky (cyl=1 , anisotropic_sky=0):

$$S_{str} = \alpha_K [K_{side,I} F_{cyl} + (K_{down} + K_{up}) F_{up} + (K_N + K_E + K_S + K_W) F_{side}] \\ + \alpha_L [(L_{down} + L_{up}) F_{up} + (L_N + L_E + L_S + L_W) F_{side}]$$

Case B — Cylinder, anisotropic sky (cyl=1 , anisotropic_sky=1):

$$S_{str} = \alpha_K [K_{side} F_{cyl} + (K_{down} + K_{up}) F_{up} + (K_N + K_E + K_S + K_W) F_{side}] \\ + \alpha_L [(L_{down} + L_{up}) F_{up} + L_{side} F_{cyl} + (L_N + L_E + L_S + L_W) F_{side}]$$

Case C — Box (cyl=0):

$$S_{str} = \alpha_K [(K_{down} + K_{up}) F_{up} + (K_N + K_E + K_S + K_W) F_{side}] \\ + \alpha_L [(L_{down} + L_{up}) F_{up} + (L_N + L_E + L_S + L_W) F_{side}]$$

Mean Radiant Temperature

$$T_{mrt} = \left(\frac{S_{str}}{\alpha_L \cdot \sigma} \right)^{1/4} - 273.2 \quad [^{\circ}\text{C}]$$

Symbol	Meaning	Default value
α_K	Human absorption coefficient for shortwave	0.70 (UI adjustable)
α_L	Human absorption coefficient for longwave	0.97 (UI adjustable)
F_{side}	Angular factor — sides (standing / seated)	0.22 / 0.167
F_{up}	Angular factor — top and bottom (standing / seated)	0.06 / 0.167
F_{cyl}	Angular factor for cylindrical direct beam (standing / seated)	0.28 / 0.20
measurement height	Height of evaluation point	1.1 m / 0.75 m

Constraint check: $F_{side} \times 4 + F_{up} \times 2 = 1$ for box; for standing cylinder: $4 \times 0.22 + 2 \times 0.06 = 1.0 \checkmark$

F-26 Wind speed height correction (PET and UTCI)

File: SOLWEIG/solweigworker.py lines 373–374

Power-law profile:

$$W_{s,PET} = \left(\frac{1.1}{z_{sensor}}\right)^{0.2} W_s$$

$$W_{s,UTCI} = \left(\frac{10}{z_{sensor}}\right)^{0.2} W_s$$

Symbol	Meaning	Unit
W_s	Station wind speed at sensor height z_{sensor}	m s ^{−1}
z_{sensor}	sensorheight from UI	m
$W_{s,PET}$	Wind speed at 1.1 m (standing person height)	m s ^{−1}
$W_{s,UTCI}$	Wind speed extrapolated to 10 m reference	m s ^{−1}

Assumption: Logarithmic/power-law neutral stability; exponent 0.2 corresponds to a roughness length appropriate for suburban terrain. This is not spatially resolved.

4. Ambiguities / Needs Verification

#	Item	Location	Status
AV-1	GVF weighting coefficients (0.5, 0.4) — (GVF1×0.5 + GVF2×0.4)/0.9 in sunonsurface_2018a.py:194 . No publication is cited in the code for these specific weights. Likely empirically tuned.	sunonsurface_2018a.py:194	Inferred from code
AV-2	TsWaveDelay decay constant 33.27 day^{−1} — This controls how quickly surface temperature follows solar	TsWaveDelay_2015a.py:11	Inferred from code

#	Item	Location	Status
	forcing. No publication cited. Equivalent to an e-folding time of ~43 min.		
AV-3	elvis correction flag — When <code>elvis=1</code> , +1 is added to <code>esky</code> (line 99). This appears to be a user-selectable emissivity correction that can push <code>esky > 1</code> , which is non-physical for a strict emissivity. Possible intended use: account for longwave emission above 1.0 that includes non-sky sources.	<code>Solweig_2022a_calc.py:99</code>	Behavior unclear; user beware
AV-4	SVF polynomial P_6 in <code>kvikt_veg.py</code> — <code>vikt_tot = 4.4897</code> and the 6th-degree polynomial coefficients are not traced to any publication in the source. They appear to be a fit to numerically integrated SVF curves.	<code>Kvikt_veg.py:5-8</code>	Inferred from code
AV-5	leafon / leafoff thresholds — Hard-coded as DOY 97 and 300 in <code>Solweig_v2015_metdata_noload.py:32-33</code> with a <code>TODO</code> comment. Users in the southern hemisphere or with non-temperate phenology will get incorrect transmissivity switching.	<code>Solweig_v2015_metdata_noload.py:32-33</code>	Known limitation, not configurable from UI
AV-6	Anisotropic longwave path for box model — When <code>cyl=0</code> and <code>anisotropic_sky=1</code> , the code adds <code>Lcyl_v2022a</code> outputs (<code>Least_</code> , <code>Lwest_</code> , etc.) to <code>Lside_veg_v2022a</code> outputs at lines 314–317. This composite is then used in the <code>Sstr</code> (Case C) formula. The physical justification for adding two independently computed longwave fields is not documented in comments.	<code>Solweig_2022a_calc.py:313-317</code>	Needs tracing to publication
AV-7	diffsh vs shmat distinction — <code>diffsh[:, :, p]</code> is used for diffuse shortwave (Perez), <code>shmat[:, :, p]</code> for building shadow patches, <code>vegshmat</code> for vegetation patches, and <code>vbshvegshmat</code> for the combined view. These are all pre-computed in <code>SkyViewFactorCalculator</code> and stored in <code>shadowmats.npz</code> . The exact generation algorithm is in a different plugin and not documented here.	<code>shadowmats.npz</code> generation in <code>SkyViewFactorCalculator</code>	External dependency
AV-8	TgOut vs Tg in POI output — The POI file writes <code>TgOut</code> (wave-delay smoothed surface temperature, col 23) rather than the raw <code>Tg</code> sinusoidal value. Users comparing POI col 23 with the diurnal wave formula (F-6) will see a smoothed version.	<code>solweigworker.py:362</code>	Documented behavior, may surprise users

#	Item	Location	Status
AV-9	CI used for nighttime Ldown cloud correction — At midnight the worker looks ahead (<code>i + rise + 1</code>) to find the first sunlit CI of the coming day and applies it retroactively to the nocturnal period. This forward-looking approach is unusual.	<code>solweigworker.py:277-283</code>	Inferred from code; no comment
AV-10	Solweig_2021a_calc.py still present — An older calculation module remains in <code>SOLWEIGpython/</code> alongside the active 2022a version. It is referenced only in commented-out code in <code>solweigworker.py</code> .	<code>SOLWEIG/SOLWEIGpython/Solweig_2021a_calc.py</code>	Dead code, safe to ignore